Supporting Information
Surface-state engineering for generation of nonlinear charge and spin photocurrents

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S1. SUPPORTING INFORMATION

S1. Expressions of charge/spin shift and injection photoconductivities

This section contains the expressions of the tensors that appear in the main text, all expressed in the length gauge as derived in Refs.[1,2].

1. *Shift*

a. Charge. The expression for the shift photoconductivity reads

$$\sigma^{abc}(\omega) = \quad -\frac{i\pi e^3}{2\hbar^2}\int \frac{d\mathbf{k}}{(2\pi)^3}\sum_{mn} f_{nm}\sigma^{abc}(\mathbf{k})\delta\left(\omega_{nm}+\omega\right). \tag{S1}$$

Here a, b and c are Cartesian components, m and n are band indexes, $f_{nm} = f_n - f_m$ is the occupation factor difference, $\omega_{mn} = \omega_m - \omega_n$ is the energy gap of the bands involved and $\sigma^{abc}(\mathbf{k}) = r^{b;a}_{mn} r^{c}_{nm} - r^{c;a}_{nm} r^{b}_{mn}$ is the $\mathbf{k}$-resolved tensor which contains interband Berry connection $r^a_{mn} = i\left\langle u_m | u_n \right\rangle$ and its generalized derivative $r^{b;a}_{mn} = \partial_a r^b_{mn} - i\left(A^a_{mm} - A^a_{nn}\right) r^b_{mn}$ with $A^a_{nn} = i\left\langle n\right| \partial_a \left|n\right\rangle$ the intraband Berry connection.

For $\mathcal{T}$-invariant systems the shift current results from the real part of the symmetric combination $\mathrm{Re}(\sigma^{abc} + \sigma^{acb})$ corresponding to the generation of current under linearly polarized light.

b. Spin. In the spin case, the expression becomes

$$\xi^{s,abc}_{shift} = \frac{i\pi q^3}{2\hbar^2 V\Omega}\sum_{\mathbf{k},m,n} f_{mn}(d^{s,ba}_{mn} v^c_{nm} - d^{s,ca}_{nm} v^b_{mn})(\mathbf{k})\delta(\Omega + \omega_{mn}) \tag{S2}$$

with $d^{s,ba} = j^{s,ab}_{mn} + \sum_{p\neq m}\frac{j^{s,a}v^b_{pn}}{\omega_{mp}} + \sum_{p\neq n}\frac{v^b_{mp}j^{s,a}_{pn}}{\omega_{np}}$, $j^{s,ab} = \frac{1}{2}\left\langle u_m\right| \left\{S^s, \partial^a\partial^b H\right\} \left|u_n\right\rangle$, $j^{s,a} = \frac{1}{2}\left\langle u_m\right| \left\{S^s, v^a\right\} \left|u_n\right\rangle$, S^s the spin operator and Ω the frequency of the incident light.

For $\mathcal{T}$-invariant systems the spin shift current results from the imaginary part of the antisymmetric combination $\mathrm{Im}(\zeta^{s,abc} + \zeta^{s,acb})$ corresponding to the generation of spin current under linearly polarized light.

2. Injection

a. Charge. The charge injection tensor can be written as

$$\eta^{abc}(\omega) = -\tau \frac{\pi q^3}{\hbar^2 \omega^2 V \Omega^2} \sum_{\mathbf{k},m,n} f_{mn} \eta^{abc}_{mn}(\mathbf{k}) \delta(\omega + \omega_{mn}), \tag{S3}$$

with $\eta^{s;abc}_{mn}(\mathbf{k}) = \Delta v^a_{mn} v^b_{mn} v^c_{nm}$ and $\Delta v^a_{mn} = v^a_{mm} - v^a_{nn}$.

For $\mathcal{T}$-invariant systems the injection current results from the imaginary part of the antisymmetric combination $\mathrm{Im}(\eta^{abc} - \eta^{acb})$ corresponding to the generation of current under circularly polarized light.

b. Spin. In the spin case, the expression becomes

$$\xi^{s,abc}(\omega) = -\tau \frac{\pi q^3}{\hbar^2 \omega^2 V \Omega^2} \sum_{\mathbf{k},m,n} f_{mn} \xi^{s,abc}_{mn}(\mathbf{k}) \delta(\omega + \omega_{mn}), \tag{S4}$$

with $\xi^{s;abc}_{mn}(\mathbf{k}) = \Delta j^{s,a}_{mn} v^b_{mn} v^c_{nm}$, $\Delta j^{s,a}_{mn} = j^{s,a}_{mm} - j^{s,a}_{nn}$ and $j^{s,a}_{mn} = \langle u_m | \{v^a, S^s\} | n \rangle$ the spin-current matrix elements as considered in[1].

For $\mathcal{T}$-invariant systems, the spin injection current results from the real part of the symmetric combination $\mathrm{Re}(\xi^{s,abc} + \xi^{s,acb})$ corresponding to the generation of spin current under linearly polarized light.

S2. Analytical expressions in the Rashba model

In this section, we provide more details on the derivation of the analytical expressions obtained for the Rashba model, as well as the complete set of charge shift and injection tensor components. The starting point is the linear Rashba Hamiltonian in polar coordinates

$$H_0 = \begin{bmatrix} \frac{\rho^2}{2m} & -i\alpha\rho e^{-i\phi} \\ i\alpha\rho e^{i\phi} & \frac{\rho^2}{2m} \end{bmatrix} \tag{S5}$$

together with an extension comprising higher order terms that comply with the C_{3v} point group

$$V = \begin{bmatrix} \beta\left(-3\rho^3\sin^2(\phi)\cos(\phi) + \rho^3\cos^3(\phi)\right) & -i\gamma\rho^3 e^{-i\phi} \\ i\gamma\rho^3 e^{i\phi} & -\beta\left(-3\rho^3\sin^2(\phi)\cos(\phi) + \rho^3\cos^3(\phi)\right) \end{bmatrix} \tag{S6}$$

which we treat as a perturbation. Up to first order in the perturbation, the eigenstates of $H = H_0 + V$ are

$$|+\rangle = \begin{bmatrix} -\frac{\sqrt{2}i\left(2\alpha+\beta\rho^2\cos(3\phi)\right)e^{-i\phi}}{4\alpha} \\ \frac{\sqrt{2}\cdot\left(2\alpha-\beta\rho^2\cos(3\phi)\right)}{4\alpha} \end{bmatrix} \tag{S7}$$

$$|-\rangle = \begin{bmatrix} \frac{\sqrt{2}i\left(2\alpha-\beta\rho^2\cos(3\phi)\right)e^{-i\phi}}{4\alpha} \\ \frac{\sqrt{2}\cdot\left(2\alpha+\beta\rho^2\cos(3\phi)\right)}{4\alpha} \end{bmatrix}, \tag{S8}$$

and the energy difference between the bands is

$$\omega_{-+} = 2\rho\left(\alpha + \gamma\rho^2\right). \tag{S9}$$

With that we solve S1 and S3, resulting in

$$\sigma^{xxx}(\mathbf{k}) = \frac{3\beta\sin(2\phi)}{2\alpha\rho} \tag{S10}$$

$$\sigma^{xxy}(\mathbf{k}) = \frac{\beta\left(16\sin^6(\phi) + 12\sin^4(\phi) - 35\sin^2(\phi) - \sin(\phi)\sin(5\phi) - 16\right)}{16\alpha\rho} \tag{S11}$$

$$\sigma^{xyy}(\mathbf{k}) = \frac{\beta\left(-16\sin^5(\phi) - 12\sin^3(\phi) + 27\sin(\phi) + \sin(5\phi)\right)\cos(\phi)}{8\alpha\rho} \tag{S12}$$

$$\sigma^{yxx}(\mathbf{k}) = \frac{\beta\left(16\sin^6(\phi) - 52\sin^4(\phi) + 21\sin^2(\phi) - \sin(\phi)\sin(5\phi) - 8\right)}{8\alpha\rho} \tag{S13}$$

$$\sigma^{yxy}(\mathbf{k}) = \frac{\beta\left(-16\sin^5(\phi) + 52\sin^3(\phi) - 13\sin(\phi) + \sin(5\phi)\right)\cos(\phi)}{16\alpha\rho} \tag{S14}$$

$$\sigma^{yyy}(\mathbf{k}) = \frac{3\beta\cos^2(\phi)}{\alpha\rho} \tag{S15}$$

and

$$\eta^{xxy}(\mathbf{k}) = \frac{-2\beta\cos(\phi)\cos(3\phi)}{4\rho^2(\alpha^2+4\alpha\gamma\rho^2)} \tag{S16}$$

$$\eta^{yxy}(\mathbf{k}) = \frac{-2\beta\sin(\phi)\cos(3\phi)}{4\rho^2(\alpha^2+4\alpha\gamma\rho^2)} \tag{S17}$$

Note that the shift (injection) photoconductivity is symmetric (antisymmetric) in $b \leftrightarrow c$, implying $\sigma^{xyx}(\mathbf{k}) = \sigma^{xxy}(\mathbf{k})$, $\sigma^{yyx}(\mathbf{k}) = \sigma^{yxy}(\mathbf{k})$, $\eta^{xyx}(\mathbf{k}) = -\eta^{xxy}(\mathbf{k})$ and $\eta^{yyx}(\mathbf{k}) = -\eta^{yxy}(\mathbf{k})$.

S3. Spin injection photoconductivity spectra of Au(111)

Fig. S1 shows the independent spin injection photoconductivity tensor components for Au(111), where it can be observed that spin-polarized in-plane components dominate over out-of-plane components. As argued in text, this is due to the in-plane spin texture which characterizes Rashba surface states.

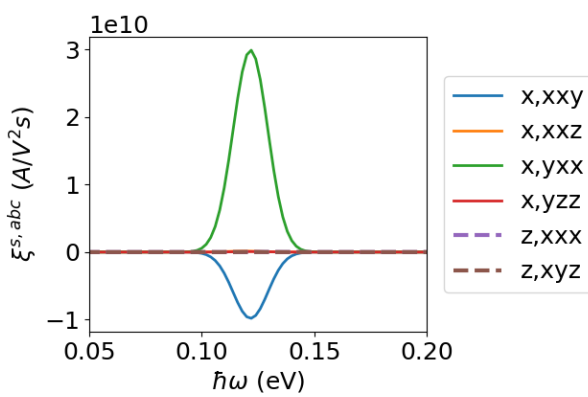

Fig. S1: Spin injection photoconductivity spectra of Au(111).

S4. Au(111) shift current spectra: Ab-initio and Rashba model

In this section we compare the calculated shift current spectra from ab-initio and the Rashba model. In order to get quantitative results in the Rashba model, we fit the parameters to reproduce Au(111) surface states energy dispersion around Γ in Fig S2a. The resulting parameters are $\alpha = 0.54$ eVÅ, $\beta = -3.13$ eVÅ^3, $\gamma = -6.25$ eVÅ^3 and $1/2m = 17.14$ eVÅ^2.

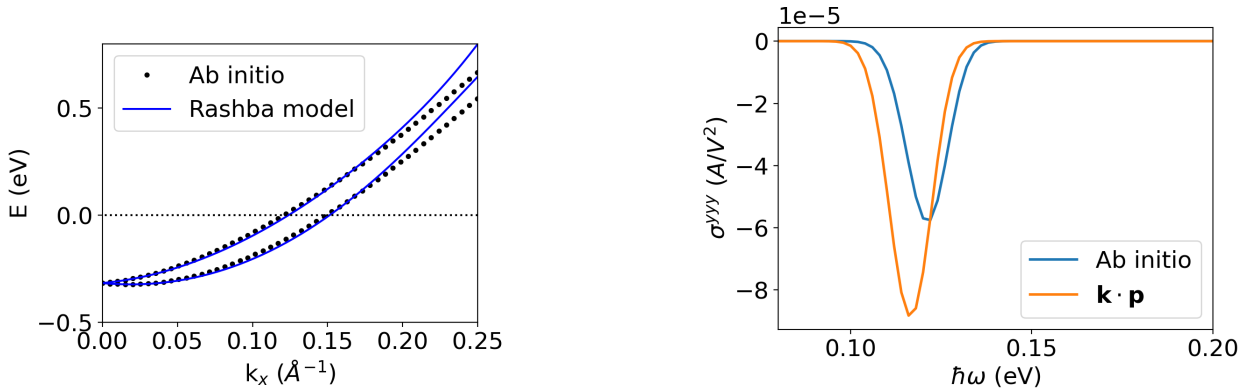

Fig. S2: a) Fitted energy dispersion around Γ with parameters $\alpha = 0.54$ eVÅ, $\beta = -3.13$ eVÅ^3, $\gamma = -6.25$ eVÅ^3 and $1/2m = 17.14$ eVÅ^2. b) Calculated shift current spectra of Au(111) by means of ab-initio and $\mathbf{k}\cdot\mathbf{p}$ methods.

Then, to calculate the shift photoconductivity spectrum we make use of the $\mathbf{k}\cdot\mathbf{p}$ functionality in the Wannierberri[3] code. Figure S2b displays the calculated spectra for both the Rashba model and ab-initio Au(111) surface states (following the same methodology as in the main text), showing a reasonable agreement as the same order of magnitude is obtained in both cases.

S5. Polarization angle dependence

The current arising from the shift mechanism can be expressed in a simple form as in Eq 1 of the main text, which reads

$$J^a(0) = \sigma^{abc}(\omega)\mathrm{Re}[E^b(\omega)E^c(-\omega)] \tag{S18}$$

We particularize for a linearly polarized electric field in the x-y plane, $\mathbf{E}(\omega) = \mathcal{E}(\omega)(\cos(\theta), \sin(\theta), 0)$ as the one discussed in the main text, Figure 2. Noting that in the case of the point group 3m, $\sigma^{yyy} = -\sigma^{yxx}$ and $\sigma^{yxy} = 0$, the current in the y direction is reduced to

$$J^y(0) = |\mathcal{E}(\omega)|^2\sigma^{yyy}(\omega)(\sin^2(\theta) - \cos^2(\theta)) = -|\mathcal{E}(\omega)|^2\sigma^{yyy}(\omega)\cos(2\theta). \tag{S19}$$

The absolute value of Eq. S19 is the quantity represented in Fig 2c of the main text, which vanishes for $\theta = 45 + 90 \cdot n$ deg.

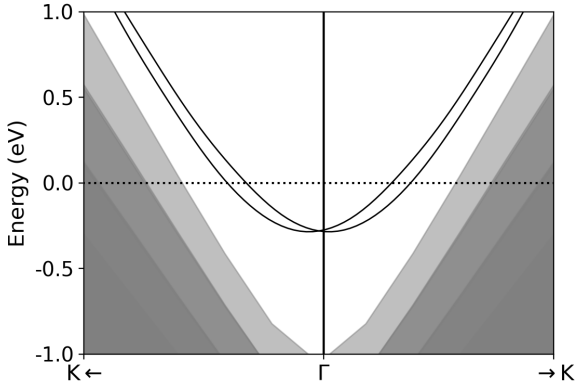

Fig. S3: Electronic band structure of Au(111) in the vicinity of the Γ point. The shaded areas denote the projected bulk bands.

We now consider a mechanism for generating a DC current through the standard photovoltaic effect. In the first place, this would involve a photoexcitation process described by the linear absorption governed by the photoconductivity tensor $\sigma^{ab}(\omega)$. Then, the photoexcited electrons would generate a DC photocurrent via the built-in electric field of the surface, in analogy with the process in pn junction solar cells. In that case, since for the 3m point group $\sigma^{yx}(\omega) = 0$ by symmetry, under $\mathbf{E}(\omega) = \mathcal{E}(\omega)(\cos(\theta), \sin(\theta), 0)$ the current along y would be proportional to $\sigma^{yy}(\omega)\sin(\theta)$, which is the angular dependence of the electrostatic contribution displayed in Fig2c of the main text. Therefore, by choosing the polarization angle as $\theta = 0 + 180 \cdot n$ deg, one can force this contribution to vanish.

S6. Wannierization parameters and band structures

a. Au(111) We constructed the Wannier model for Au(111) using a selective approach where only the two surface states crossing the Fermi level were considered, and the rest were excluded. We employed a spinor p_z orbital located at the position of the top-most Au atom as initial projections The calculated band structure, shown in Figure S3, displays the Wannier-interpolated electronic bands near the Γ point, where the surface states can be distinguished from the bulk bands.

b. Tl/Si(111) To model the unoccupied states of the Tl/Si(111) system (see Figure 2a in the main text), we considered a wannierization procedure that contains only those bands, excluding the rest. We used a spinor p_z orbital located at the position of the Tl atom as

initial projections.

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